YIYOU SUN

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Education

University of Wisconsin-Madison Sept. 2018 - July. 2023
PhD. in Computer Science, Minor in Math *Madison, US*

- GPA 3.85/4.0. Advised by Prof. Sharon (Yixuan) Li.

Harbin Institute of Technology Sept. 2014 - Jun. 2018
B.S. in Computer Science *Harbin, China*

- GPA 93.67/100. Rank **1st** among 97 students.

Research Overview

– *I aim to pave the way to a reliable AI system in generating controlled and predictable content.*

Current Research Topics: Large-language Model (Dialogue System, Multi-modality, Privacy, Uncertainty Estimation, Trustworthiness).

Ph.D. Research Topics: Out-of-distribution (OOD) Detection, Interpretability, (Generalized) Novel Class Discovery, Representation Learning.

Academic Experience

University of Wisconsin-Madison Sept 2020 - July 2023
Research Assisstant *Madison, WI*

- Advised by Prof. Sharon (Yixuan) Li.
- Finished thesis "*Detecting and Learning Out-of-Distribution Data in the Open-world: Algorithm and Theory*", investigating two essential steps for open-world machine learning: Out-of-distribution (OOD) Detection and Open-world Representation Learning (ORL).

University of Wisconsin-Madison Sept 2018 - May 2020
Research Assisstant *Madison, WI*

- Advised by Prof. Vikas Singh.

MIT: Computer Science and Artificial Intelligence Laboratory Dec 2017 - May 2018
Undergraduate Research Assisstant *Cambridge, MA*

- Supervised by Dr. Bolei Zhou (now at UCLA).

Harvard University: DASlab July 2017 - May 2018
Undergraduate Research Assisstant *Cambridge, MA*

- Supervised by Prof. Stratos Idreos.

Industrial Experience

NEC Laboratories America Inc. Aug. 2023 - Current
Researcher *Plainsboro, NJ*

Google Research May 2022 - Sept 2022
Research Intern *Mountain View, CA*

Facebook Integrity Team June 2021 - Sept 2021
Research Intern *Remote*

Facebook AI June 2020 - Sept 2020
Research Intern *Remote*

Publications (Sorted by Years)

(* indicates equal contribution)

[1] **Y. Sun**, J. Hu, W. Cheng, and H. Chen. "Chatbot Meets Pipeline: Augment Large Language Model with Definite Finite Automaton." In Proceedings of the International Conference on Machine Learning (ICML), 2024.

[2] X. Du, **Y. Sun** and Y. Li. "When and How Does In-Distribution Label Help Out-of-Distribution Detection?" In Proceedings of the International Conference on Machine Learning (ICML), 2024.

[3] L. Chen, X. Zhao, W. Cheng, Y. Liu, **Y. Sun**, X. Zhang, M. Oishi, T. Osaki, K. Matsuda, J. Ji, G. Bai, L. Zhao, H. Chen. "Uncertainty Decomposition and Quantification for In-Context Learning of Large Language Models." 2024 Annual Conference of the Association for Computational Linguistics (NAACL), 2024.

[4] S. Ghosal*, **Y. Sun***, Y. Li. "*How to Overcome Curse-of-Dimensionality for OOD Detection?*". In the 38th AAAI Conference on Artificial Intelligence (AAAI), 2024.

[5] M. Longfei*, **Y. Sun***, K. Ding, and F. Wu. "Score Propagation as a Catalyst for Graph Out-of-distribution Detection: A Theoretical and Empirical Study." OpenReview, (2023).

[6] (**Spotlight**) **Y. Sun**, S. Zhenmei, Y. Li. "*A Graph-Theoretic Framework for Understanding Open-World Semi-Supervised Learning*". In Proceedings of the Advances in Neural Information Processing Systems (NeurIPS), 2023.

[7] X. Du, **Y. Sun**, X. Zhu, Y. Li "*Dream the Impossible: Outlier Imagination with Diffusion Models.*" In Proceedings of the Advances in Neural Information Processing Systems (NeurIPS), 2023.

[8] Z. Jingyang, J. Yang, P. Wang, H. Wang, Y. Lin, H. Zhang, **Y. Sun**, X. Du, K. Zhou, W. Zhang, Y. Li, Z. Liu, Y. Chen, H. Li. "OpenOOD v1.5: Enhanced Benchmark for Out-of-Distribution Detection". arXiv preprint arXiv:2306.09301, 2023.

[9] **Y. Sun**, Y. Liu, X. Zhu, Y. Li, W. Chu. "*Rethinking Domain Generalization for Face Anti-spoofing: Separability and Alignment*". In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2023.

[10] Y. Ming, **Y. Sun**, O. Dia, and Y. Li. "*How to Exploit Hyperspherical Embeddings for Out-of-Distribution Detection?*." International Conference on Learning Representations (ICLR), 2023.

[11] **Y. Sun**, S. Zhenmei, L. Yingyu, Y. Li. "*When and How Does Known Class Help Discover Unknown Ones? Provable Understanding Through Spectral Analysis*". In Proceedings of the International Conference on Machine Learning (ICML), 2023.

[12] Y. Ming, Z. Cai, J. Gu, **Y. Sun**, W. Li, Y. Li. "*Delving into OOD Detection with Vision-Language Representations.*" In Proceedings of the Advances in Neural Information Processing Systems (NeurIPS), 2022.

[13] J., Yang, P. Wang, D. Zou, Z. Zhou, K. Ding, W. Peng, H. Wang, G. Chen, B. Li, **Y. Sun**, X. Du, K. Zhou, W. Zhang, D. Hendrycks, Y. Li, Z. Liu. "*OpenOOD: Benchmarking Generalized Out-of-Distribution Detection.*" In Proceedings of the Advances in Neural Information Processing Systems (NeurIPS), 2022.

[14] **Y. Sun**, Y. Ming, X. Zhu, Y. Li. "*Out-of-distribution Detection with Deep Nearest Neighbors*". In Proceedings of the International Conference on Machine Learning (ICML), 2022.

[15] **Y. Sun**, Y. Li. "*Dice: Leveraging sparsification for out-of-distribution detection.*" In Proceedings of the the European Conference on Computer Vision (ECCV), 2022.

[16] **Y. Sun**, Y. Li. "*OpenCon: Open-world Contrastive Learning with Wild Unlabeled Data*". Transactions on Machine Learning Research (TMLR), 2022.

[17] **Y. Sun**, C. Guo, Y. Li. "*ReAct: Out-of-distribution Detection with Rectified Activations.*" In Proceedings of the Advances in Neural Information Processing Systems (NeurIPS), 2021.

[18] **Y. Sun**, B. Joseph, A. Deatsh, R. Jeraj, Y. Li. "*LOOD: Localization-based Uncerntainty Estimation for Medical Imaging.*" ICML Workshop on Distribution-Free Uncertainty Quantification (ICML DFUQ), 2021.

[19] **Y. Sun**, S. Ravi, V. Singh. "*Adaptive Activation Thresholding: Dynamic Routing Type Behavior for Interpretability in Convolutional Neural Networks.*" In Proceedings of the IEEE Conference on Computer Vision (ICCV), 2019.

[20] B. Zhou*, **Y. Sun***, D. Bau*, A. Torralba. "*Interpretable Basis Decomposition for Visual Explanation.*" In Proceedings of the European Conference on Computer Vision (ECCV), 2018. (* indicates equal contribution)

[21] B. Zhou, **Y. Sun**, D. Bau, A. Torralba. "*Revisiting the Importance of Individual Units in CNNs via Ablation*". arXiv preprint arXiv:1806.02891, 2018.

[22] S. Idreos, K. Zoumpatianos, M. Athanassoulis, N. Dayan, B. Hentschel, M. S. Kester, D. Guo, L. M. Maas, W. Qin, A. Wasay, and **Y. Sun**. "*The Periodic Table of Data Structures*". IEEE Data Eng. Bull., 41(3):64-75, 2018.

[23] S. Idreos, K. Zoumpatianos, S. Chatterjee, W. Qin, A. Wasay, B. Hentschel, M. Kester, N. Dayan, D. Guo, M. Kang, **Y. Sun**. "*Learning Data Structure Alchemy*". Bull. of the IEEE Computer Society Technical Committee on Data Engineering. 42(2), 2018.

[24] **Y. Sun**, T. Su, Z. Tu. "*Faster R-CNN Based Autonomous Navigation for Vehicles in Warehouse*". In 2017 IEEE International Conference on Advanced Intelligent Mechatronics (AIM), pages 1639-1644, July 2017.

Community Services

Journal Reviewer

- Review for Pattern Recognition in 2021, 2022.
- Review for IEEE Transactions on Cybernetics in 2022.
- Review for IEEE Transactions on Multimedia in 2023.
- Review for International Journal of Computer Vision (IJCV) in 2022, 2023.
- Review for IEEE Transactions on Image Processing (TIP) in 2023.

Conference Reviewer

- Review for Conference on Computer Vision and Pattern Recognition (CVPR) in 2022, 2023, 2024.
- Review for International Conference on Machine Learning (ICML) in 2022, 2023, 2024.
- Review for Conference on Neural Information Processing Systems in 2022 (**Top Reviewer Award**), 2023.
- Review for European Conference on Computer Vision (ECCV) in 2022.
- Review for International Conference on Learning Representations (ICLR) in 2022, 2023, 2024.
- Review for Association for the Advancement of Artificial Intelligence (AAAI) in 2022, 2023.

Talks

Introduction on Trustworthy Machine Learning Nov 2023
Guest lecture @ UC Riverside CS/ECE 228 – Introduction to Deep Learning *Riverside, CA*

The Theoretical Foundation for the Open-world Representation Learning: A Spectral View Oct 2023
Invited Talk @ UIUC ML Seminar *Remote*

Out-of-distribution (OOD) Generalization and Detection in AI Risk Management Oct 2023
Invited Talk @ Beijing Academy of Artificial Intelligence *Remote*

Detecting and Learning Out-of-Distribution Data in the Open world July 2023
Ph.D. Oral Defense *Madison, WI*

Rethinking Domain Generalization for Face Anti-spoofing: Separability and Alignment Sept 2022
Intern Presentation @ Google Research *Mountain View, CA*

Introduction on Interpretable Deep Learning Oct 2020
Guest lecture @ CS 839 – Special Topics in AI: Deep Learning *Madison, WI*